

Extended Kalman Filter Tuning for speed and torque estimation of an AC
induction motor using U-NSGA-III multi-objective genetic algorithm

Term project report

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Abstract

For achieving precise speed-sensor-less control of an AC induction motor (IM), AC drives require accurate estimations of speed and electrical torque of the motor. Space vector equations based mathematical models of the IM are generally employed to obtain the speed and other variable information using the measurement of voltage and current at the IM terminals. Extended Kalman filter (EKF) is an effective estimation technique which is employed for this. As it is stochastic in nature, has high convergence rate, and good disturbance rejection. However, it relies on valid selection for its covariance filter matrices Q and R . Traditionally, EKFs are tuned by selecting Q and R such that a single estimation parameter (for example rotor speed) is optimized. Multi-objective genetic algorithms can be used to concurrently achieve optimal torque and speed estimations which in-turn improves the performance of the complete control scheme of the AC drive. This project explores the use of unified non-dominated sorting genetic algorithm-III (U-NSGA-III) for achieving this task. By using U-NSGA-III to tune Q and R matrices for the EKF estimator, better state estimates of torque and speed are expected, resulting in better low-speed torque control and set point tracking. The proposed optimizing technique for EKF-based estimation scheme is used in the combination with the sensor less direct torque control of induction motor. The multi-optimal based-EKF is tested with three regions of Pareto front curve.

1. Introduction

Induction motors are the most commonly and widely used electrical machines in several industrial applications, they are cheaper, rugged, and easier to maintain compared to other alternatives. However, its non-linear structure requires a decoupled torque and flux control. The field orientation control, proposed by F. Blaschke in 1972, is a technique that is used to control the induction motor speed, this technique allows a decoupling between torque and flux in order to obtain independent components control exactly like DC motors.

The three-phase inverters employed in the AC drives which are used in space vector control of induction motor are traditionally controlled by the triangulo-sinusoidal pulse width modulation. But this technique requires a speed sensor for rotor angular speed measurement. This technique has another major disadvantage, which is the increase in switching loss per period and reduction in opening-closing switches cycle, so it requires better switching performance.

Alternatively, speed sensor-less control methods of induction motors drive using the estimated speed instead of the measured one. The speed has been estimated from the instantaneous values of

stator voltages and currents using the induction motor dynamic model in field rotating reference frame.

The EKF is a popular estimation technique due to its stochastic nature, high convergence rate, and good disturbance rejection capabilities. However, it is known that the success of the EKF relies on the correct selection of covariance filter matrices. As a result, there have been several research efforts focused on optimizing the EKF for single parameter estimation in AC drives using different optimization techniques. However, few studies have focused on optimizing two state parameters simultaneously.

This project uses the unified multi-objective non-dominated sorting genetic algorithm-III, UNSGA-III, algorithm to concurrently optimize the EKF covariance matrices for optimal torque and speed estimation, leading to improved overall system performance. Traditional optimization schemes become more complex when optimizing multiple variables, as optimizing one parameter can affect the other. By using multi-objective optimization, the trade-off between the errors of the estimated states (ω_r , τ_e) can be resolved, resulting in higher accuracy, and improved dynamic performance of electrical drives. It is worth noting that torque estimation is often performed by calculating the torque from four estimated IM variables, so the estimated values of these variables are directly influenced by the estimation of torque. As a result, the overall system performance will be enhanced when the estimated speed and torque are concurrently optimized by the proposed algorithm.

2. Induction motor modelling

In this project, the mathematical model of a three-phase squirrel cage induction motor is described in the rotor fixed reference frame (d, q) by assuming the stator currents and the rotor fluxes as state variables:

$$\begin{aligned} \dot{x} &= \mathbf{A}.x + \mathbf{B}.u \\ y &= \mathbf{C}.x \end{aligned} \quad (1)$$

$$\begin{aligned} x &= [i_{sd} \ i_{sq} \ \phi_{rd} \ \phi_{rq}]^t ; u = [u_{sd} \ u_{sq}]^t ; y = [i_{sd} \ i_{sq}]^t \\ \mathbf{A} &= \begin{bmatrix} -\gamma & \omega_s & \frac{K}{T_r} & K\omega_r \\ \omega_s & -\gamma & -\frac{K}{T_r} & \frac{K}{T_r} \\ \frac{L_m}{T_r} & 0 & -\frac{1}{T_r} & \omega_{sl} \\ 0 & \frac{L_m}{T_r} & -\omega_{sl} & -\frac{1}{T_r} \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} \frac{1}{\sigma L_s} & 0 \\ 0 & \frac{1}{\sigma L_s} \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \\ \mathbf{C} &= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \end{aligned}$$

With:

$$\gamma = \frac{R_s}{\sigma L_s} + \frac{1-\sigma}{\sigma T_r} ; K = \frac{1-\sigma}{\sigma L_m} ; \sigma = 1 - \frac{L_m^2}{L_s L_r} ; T_r = \frac{R_r}{L_r}$$

where:

(U_{sd}, U_{sq}) , (I_{sd}, I_{sq}) and (I_{rd}, I_{rq}) , are Stator voltage, Stator current and rotor current space vectors respectively.

(Φ_{rd}, Φ_{rq}) are rotor flux linkage space vectors.

L_s , L_r are Stator and rotor self-inductances.

R_s , R_r are Stator and rotor resistances. And

ω_r is the Rotor speed.

The electromagnetic torque can be calculated using estimated state variables which are the stator current and rotor flux by:

$$\hat{T}_{ei} = \frac{3}{2} \frac{p}{2} \frac{L_m}{L_r} (i_{sq} \hat{\Phi}_{dr} - i_{sd} \hat{\Phi}_{qr}) \quad (2)$$

For the induction motor, following parameters are selected in the project:

Parameter	Rated Value
Power	200 hp
Voltage	460 V (rms 3ph)
Frequency	60 Hz
Pair pole	4
Rated speed	1785 RPM
Stator resistance R_s	3 Ω
Rotor resistance R_r	4.1 Ω
Stator inductance L_s	0.5846 H
Rotor inductance L_r	0.5937 H
Mutual inductance L_m	0.5664 H
Moment of inertia of rotor	0.0031 Kg. m ²
Viscous friction co-efficient	0 N s/rad

3. The extended Kalman Filter

The extended Kalman filter estimates the state of a non-linear process. Particularly, it allows to add, to the state vector, another variable that is needed to be estimated. The EKF is commonly used for estimating various quantities in induction machines, such as rotor speed, load torque, and electrical and mechanical parameters. Because the EKF is simply the application of the standard Kalman filter to a non-linear system, it is necessary to perform a linearization of the system at each step around the operating point. This is achieved by using the following non-linear model of the system:

$$\begin{aligned} X_{e_{k+1}} &= f(X_{e_k}, U_k) + W_k \\ Y_k &= h(X_{e_k}) + V_k \end{aligned} \quad (3)$$

Such that:

$$X_{e_k} = [X_k \ \Theta_k]^t \quad (4)$$

Here f and h are non-linear functions, X_{ek} represents the extended state vector, where X_k and Θ_k are considered the main state vector and the estimated parameter vector (made up of the parameters that are needed to be estimated). The parameters that are required to be estimated are such that they vary quite less compared to the main state vectors. Hence their dynamics are written as: $\Theta_k = \Theta_{k+1}$.

This results in the following augmented state model:

$$\begin{aligned} \begin{bmatrix} X_{k+1} \\ \Theta_{k+1} \end{bmatrix} &= \begin{bmatrix} A_d & 0 \\ 0 & I \end{bmatrix} \begin{bmatrix} X_k \\ \Theta_k \end{bmatrix} + \begin{bmatrix} B_d \\ 0 \end{bmatrix} U_k + W_k \\ Y_k &= [C_d \ 0] \begin{bmatrix} X_k \\ \Theta_k \end{bmatrix} + V_k \end{aligned} \quad (5)$$

Where A_d , B_d and C_d are the state, the input and the discrete output matrices, respectively. I is the identity matrix. As the Kalman filter a discrete time observer, A_d , B_d and C_d are obtained by discretizing (the sampling rate T_s is set to 100 μs in simulation) the A , B and C matrices defined in equations (1). For a three-phase induction machine model described by equations (1), they can be written as:

$$A_d \approx e^{A \cdot T_s} = I + A \cdot T_s; B_d = B \cdot T_s; C_d = C.$$

$$X_e = [X \quad \Theta]^t = [i_{s\alpha} \quad i_{s\beta} \quad \phi_{r\alpha} \quad \phi_{r\beta} \quad \omega_r]^t \quad Y = [i_{s\alpha} \quad i_{s\beta}]^t \quad U = [u_{s\alpha} \quad u_{s\beta}]^t$$

With:

$$A_d = \begin{bmatrix} 1 - T_s \cdot \gamma & 0 & T_s \cdot \frac{\mu}{T_r} & T_s \cdot \mu \cdot \omega_r & 0 \\ 0 & 1 - T_s \cdot \gamma & -T_s \cdot \mu \cdot \omega_r & T_s \cdot \frac{\mu}{T_r} & 0 \\ T_s \cdot \frac{L_m}{T_r} & 0 & 1 - \frac{T_s}{T_r} & -T_s \cdot \omega_r & 0 \\ 0 & T_s \cdot \frac{L_m}{T_r} & T_s \cdot \omega_r & 1 - \frac{T_s}{T_r} & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}; B_d = \begin{bmatrix} T_s \cdot \frac{1}{\sigma \cdot L_s} & 0 \\ 0 & T_s \cdot \frac{1}{\sigma \cdot L_s} \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \quad (6)$$

To implement the extended Kalman filter algorithm on a discrete nonlinear system the execution of the following steps is required:

- Initialize the states $X_{0|0}$, the parameters $\Theta_{0|0}$ and the covariance matrices $P_{0|0}$, Q and R
- Predict the states and the parameters at next $(k+1)$ instant

$$X_{k+1|k} = A_d \cdot X_{k|k} + B_d \cdot U_k$$

$$\Theta_{k+1|k} = \Theta_{k|k}$$

- Predict the error covariance matrix P

$$P_{k+1|k} = F_k \cdot P_{k|k} \cdot F_k^t + Q$$

Where,

$$F_k = \begin{bmatrix} A_d & \frac{\partial}{\partial \Theta_k} (A_d \cdot X_{k|k} + B_d \cdot U_k) \Big|_{\Theta_{k|k}} \\ 0 & I \end{bmatrix}$$

- Calculate Kalman gain

$$K_{k+1} = P_{k+1|k} \cdot H_k^t + [H_k \cdot P_{k+1|k} \cdot H_k^t + R]^{-1}$$

Where,

$$H_k = \begin{bmatrix} C_d & \frac{\partial}{\partial \Theta_k} (C_d \cdot X_{k|k}) \Big|_{\Theta_{k|k}} \\ 0 & I \end{bmatrix}$$

- Estimate the states and the parameters

$$\begin{bmatrix} X_{k+1|k+1} \\ \Theta_{k+1|k+1} \end{bmatrix} = \begin{bmatrix} X_{k+1|k} \\ \Theta_{k+1|k} \end{bmatrix} + K_{k+1} \cdot [Y_{k+1} - C_d \cdot X_{k+1|k}]$$

- Estimate the error covariance matrix

$$P_{k+1|k+1} = P_{k+1|k} - K_{k+1} \cdot K_k \cdot P_{k+1|k}$$

- Update the time instant to $k+1$ and repeat from the second step.

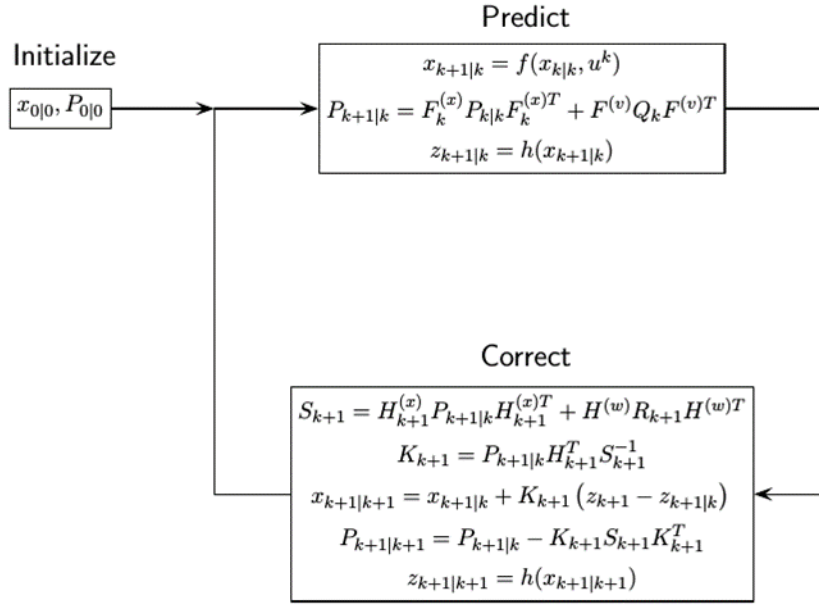


Figure 1

4. Unified multi-objective non-dominated sorting genetic algorithm-III (UNSGA-III)

The U NSGA-III algorithm is similar to the NSGA-III, except, instead of selecting parents randomly for mating, it uses tournament selection. The U stands for unified and it increases NSGA-III's performance by introducing tournament pressure.

For NSGA-III and thus also for UNSGA-III, first, the non-dominated sorting is done similar to NSGA-II. Wherein, after the population is initialized, Sorting process based on non-domination criteria of the population is done. However, unlike in NSGA-II, a series of reference points are introduced to improve the diversity and convergence. These reference points are placed in a normalized hyper-plane using Das and Dennis's systematic approach [1]. The NSGA-III uses a set of reference directions to maintain diversity among solutions. A reference direction is a ray starting at the origin point and passing through a reference point.

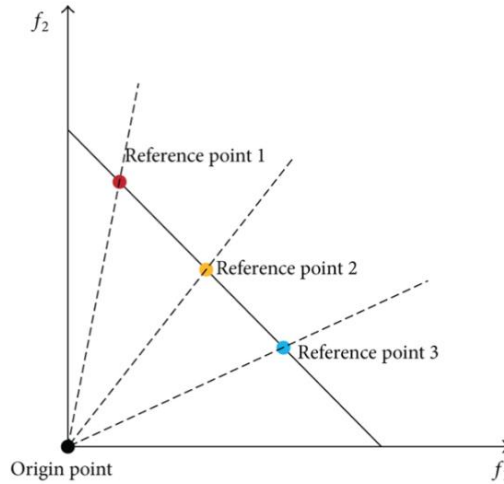


Figure 2

The population size is chosen in smallest (4) multiples, such that for every reference direction, one population member is expected to be found. The offspring population having N members is obtained by recombination and mutation after tournament selection. To preserve elite members, the combination of parents and offspring are sorted to different nondomination levels. Thereafter, individuals of each nondomination level are selected to construct a new population. The last level individuals are only sorted partially by the diversity maintenance operator. Before the previous operations, objective values are normalized. After the normalizing operation, the original reference points calculated lie on this normalized hyperplane. In order to associate each population member with a reference point, the shortest perpendicular distance between each individual of new population and each reference line is calculated. Finally, a niche-preservation strategy is employed to select individuals from top level that are associated with each reference point.

The non-dominated sorting of individuals in population is done based on rank and crowding distance.

In the algorithm used in the project, SBX crossover and polynomial mutation is employed.

The following is the flow-chart for the UNSGA – III Algorithm.

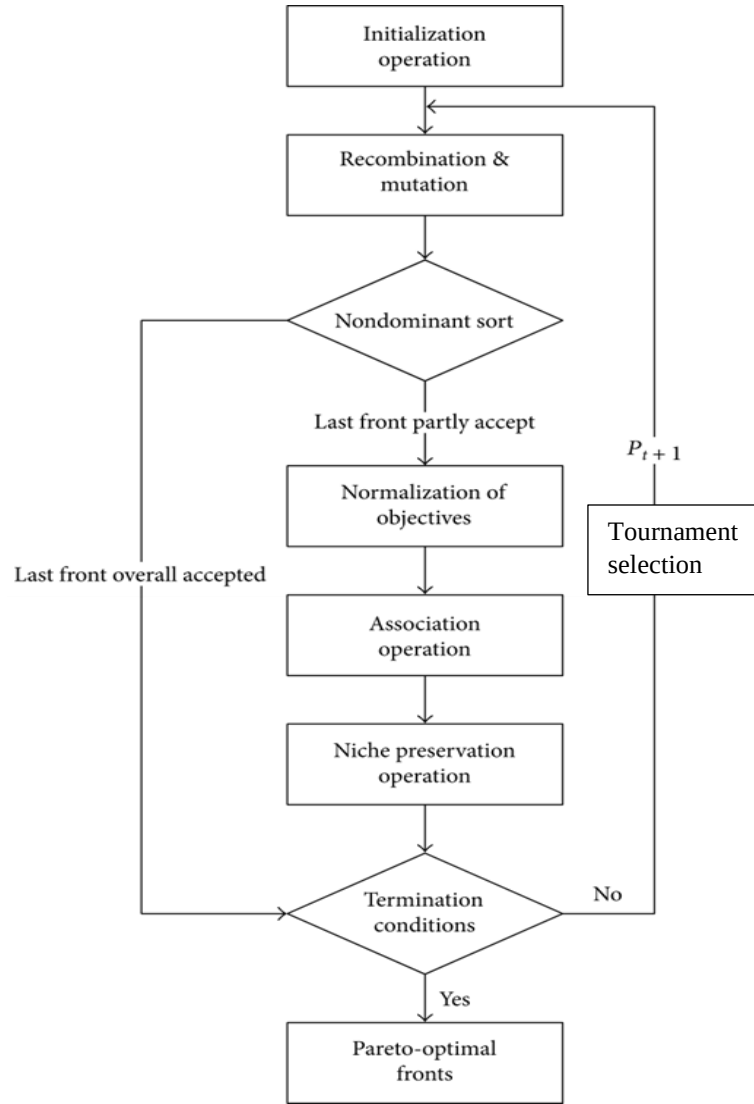


Figure 3

5. Problem formulation

The objective of the project is to estimate the rotor speed ω_r and torque τ_r of an induction motor without direct measurement using sensors. Thus, these entities are needed to be estimated from stator voltages and current. As stated prior the EKF estimation technique is employed for this.

In the EKF estimation technique, there are two matrices Q and R that are needed to be optimized. For this aim, a multi-objective optimization method UNSGA III is implemented.

The Population representation of the decision variables which implies the five diagonal elements (q_{11} , q_{22} , q_{33} , q_{44} , q_{55}) of the matrix Q , and two diagonal elements (r_{11} , r_{22}) of covariance matrix R are coded into a long string, known as an individual.

$$Q = \begin{bmatrix} q_{11} & 0 & 0 & 0 & 0 \\ 0 & q_{22} & 0 & 0 & 0 \\ 0 & 0 & q_{33} & 0 & 0 \\ 0 & 0 & 0 & q_{44} & 0 \\ 0 & 0 & 0 & 0 & q_{55} \end{bmatrix}, \quad R = \begin{bmatrix} r_{11} & 0 \\ 0 & r_{22} \end{bmatrix}$$

Thus, $individual = [q_{11}, q_{22}, q_{33}, q_{44}, q_{55}, r_{11}, r_{22}]$.

Using the Pymoo python library the individual are selected from a search space of $[1e-30, 1e+50]$ for all the decision variables. The parameters for the UNSGA III algorithm are:

- Population size = 100
- Number of generations = 20
- Sampling = Float Random sampling
- Selection = Tournament selection
- Crossover operator = SBX
- Mutation operator = Polynomial mutation
- Number of Reference directions = 20

As the rotor speed ω_r and torque τ_r are required to be estimated, the two objectives to be minimized are: the mean-squared error of the estimated speed, given by $f_{1(x)}$, and mean-squared error of the estimated torque, given by $f_{2(x)}$.

$$\begin{aligned} f_1(x) &= \frac{1}{n} \sum_{i=1}^n (\omega_{ri} - \hat{\omega}_i)^2 \\ f_2(x) &= \frac{1}{n} \sum_{i=1}^n (te_{mi} - \hat{te}_i)^2 \end{aligned} \quad (7)$$

Where $\hat{\omega}_r$ is the estimated speed and ω_r is the actual speed. Similarly, $\hat{te}_e$ is the estimated electromagnetic torque and te_e is the real torque. Note that te_e is not estimated directly, but is calculated from the state estimates of the stator current and rotor flux of the EKF, by using the formula (2), stated prior.

To tune said EKF estimator, training data of input – output pairs is required. The input being stator voltages and currents, and the output being the rotor torque and speed. For this a non-linear model of the induction machine is simulated in MATLAB Simulink. The nonlinear model is based on the d-q axis equivalent circuit of an induction machine described prior. The Simulink model is illustrated in figure 4.

The Simscape (electrical) specialized power systems library was used to create the model.

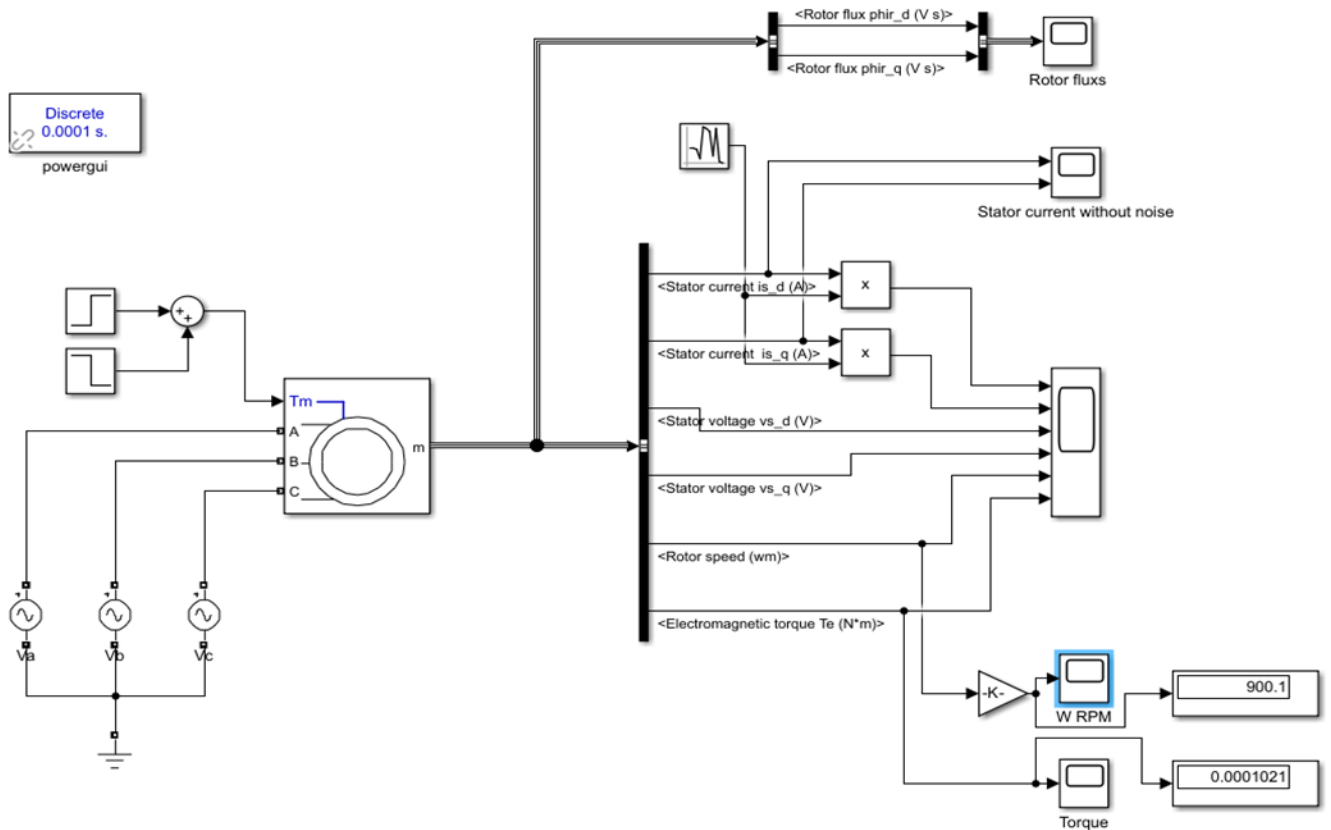


Figure 4

The simulation is run in a discrete time step of 100 μ s, as this was the assumed sampling rate for the EKF. The reference rotor speed is set to 1500 RPM. The simulation is ran for 4 seconds yielding 4001 vectors of input-output pairs and state vectors. The head of the data frame is illustrated under:

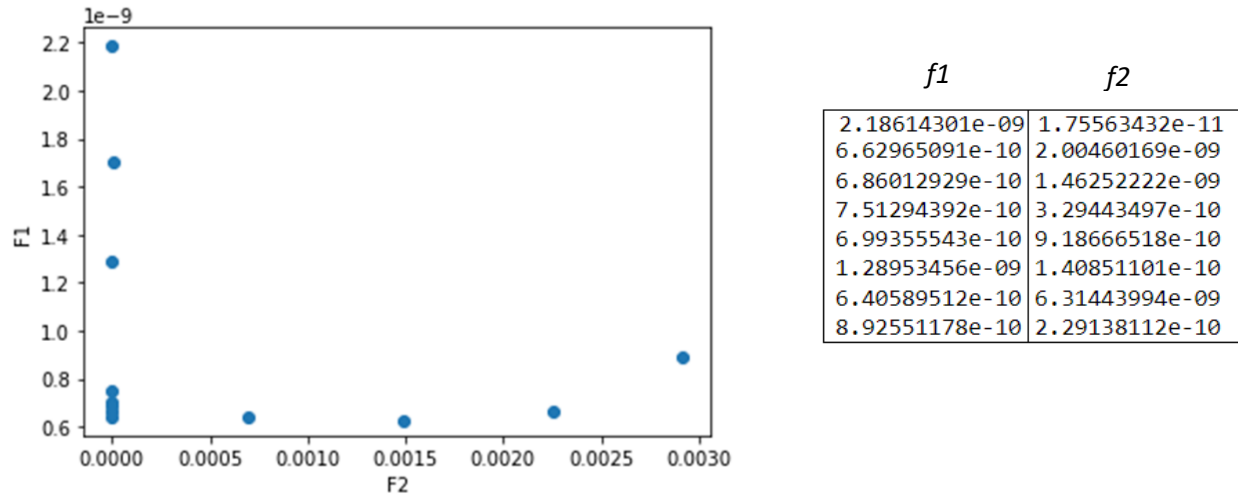
	time	id_s	iq_s	vd_s	vq_s	W	T
0	0.0000	0.000000	0.000000	375.588427	0.000000	94.247780	0.000000e+00
1	0.0001	0.000000	0.000000	375.321561	14.155996	94.247780	0.000000e+00
2	0.0002	0.043426	0.000819	374.521343	28.291876	94.247773	-1.020000e-08
3	0.0003	0.086770	0.003273	373.188909	42.387552	94.247770	-8.140000e-08
4	0.0004	0.129971	0.007358	371.326154	56.422993	94.247767	-3.350000e-07

The measured scope output of rotor speed and torque, along with the input stator current and stator voltage in Clark and Park transformed d-q frame, are saved and exported as a CSV file. This CSV file is subsequently imported to a Jupyter notebook, where the EKF algorithm function as well as the pymoo UNSGA-III algorithm are set up. And the individual consisting of Q and R matrix entries [q11, q22, q33, q44, q55, r11, r22] is optimized. A pareto optimal front is obtained

from this and the best trade-off solutions suggested by the UNSGA III algorithm are selected and the tracking ability of the estimator is tested by simulating with a step input reference.

6. Simulation results

The pareto optimal front obtained from running the UNSG – III algorithm is as following:

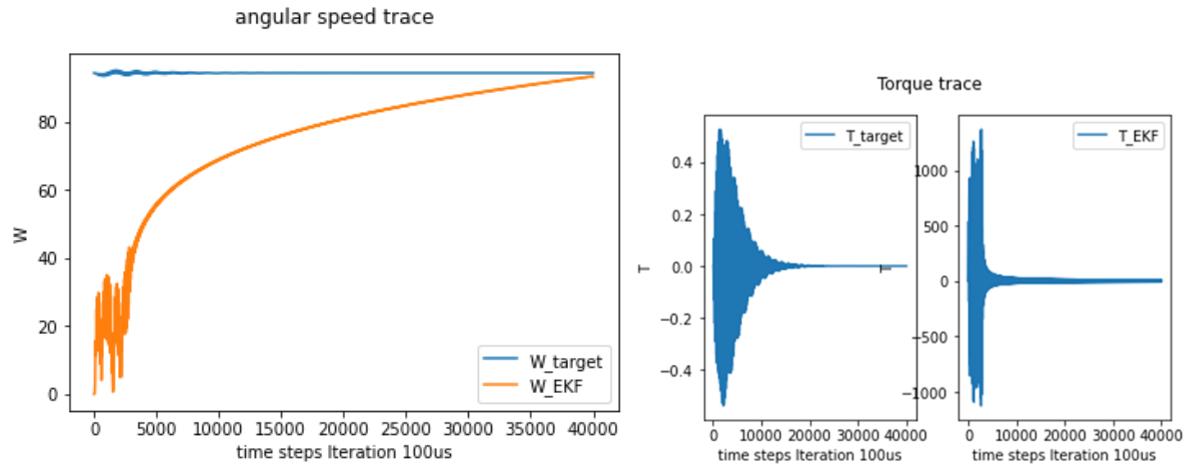


The corresponding values for the Q and R matrix are provided below in sequence of [q11, q22, q33, q44, q55, r11, r22].

```
X = [[1.08679656e+35 6.19095302e+37 4.15123986e+37 5.27568237e+37
1.40547465e+37 3.64803647e+37 9.01442191e+37]
[1.56890257e+36 1.28867695e+35 2.50297148e+37 5.50162672e+37
9.67063259e+37 6.38441194e+37 8.64976448e+37]
[1.32377400e+37 3.75511204e+35 2.37827572e+37 5.00314666e+37
8.87660293e+37 5.08731027e+37 7.47266186e+37]
[2.28273541e+35 1.73935355e+35 2.37807644e+37 5.05723520e+37
6.96667252e+37 6.30769261e+37 8.25465082e+37]
[1.97396324e+36 2.13577883e+35 2.50297148e+37 6.02615796e+37
9.19680632e+37 6.38441194e+37 8.42459469e+37]
[2.05243474e+35 2.00486806e+37 9.73489160e+37 5.52796505e+37
1.63205338e+37 9.44086511e+37 7.72360347e+37]
[1.92615834e+36 2.14353463e+35 2.50297148e+37 4.11065669e+37
9.48702371e+37 5.29541465e+37 8.25465082e+37]
[1.32414459e+37 3.95438063e+35 2.46811733e+37 5.51038020e+37
6.31709576e+37 6.12697376e+37 8.25471120e+37]]
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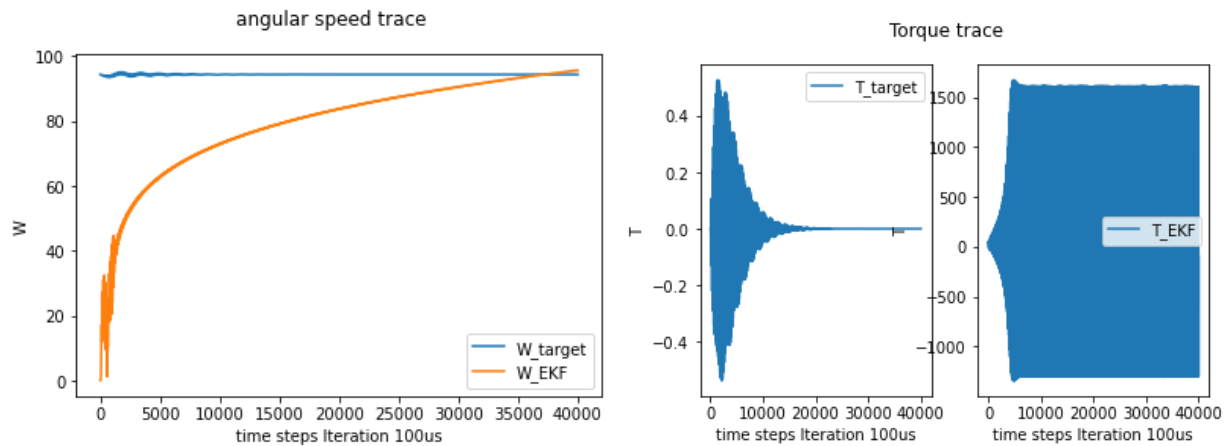
Estimates of speed and torque given by the EKF for three different selections of Q and R matrix values is provided under:

For Q = ([-1.97396324e+36, -2.13577883e+35, -2.50297148e+37, -6.02615796e+37, -9.19680632e+37]) and R = ([-6.38441194e+37, -8.42459469e+37]), which is a good trade off solution the estimations are given by the following plots:



It can be noted that, this solution provides good tracking for both rotor torque and speed.

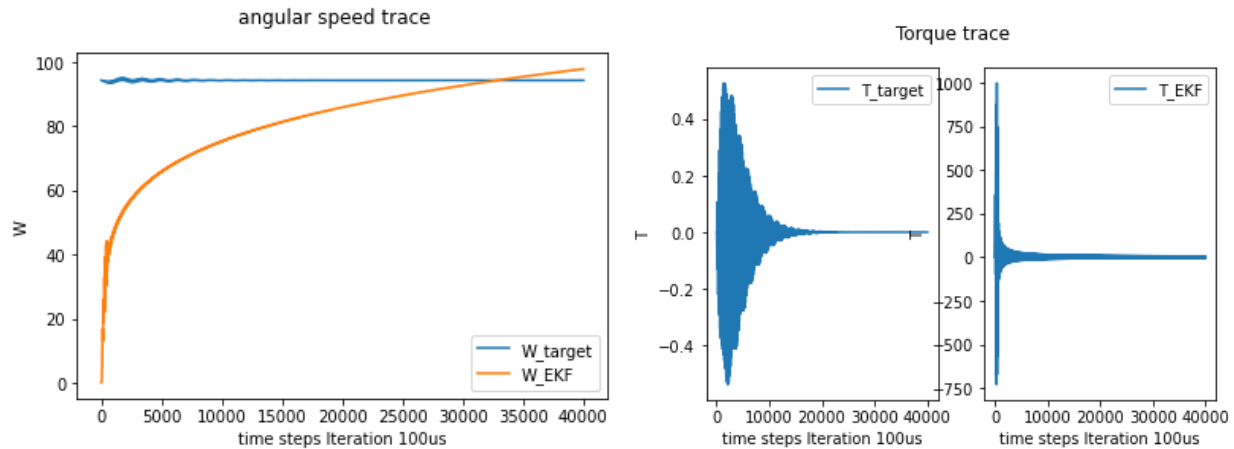
For $Q = ([-1.56890257e+36, -1.28867695e+35, -2.50297148e+37, -5.50162672e+37, -9.67063259e+37])$ and $R = ([-6.38441194e+37, -8.64976448e+37])$, which gives the full weightage to angular speed, the estimations are given by the following plots:



This solution provides good rotor speed tracking but poor torque estimation.

Note that the reference torque units in the graph are N-m and that of etimated torque is Kg – mm. Hence, the discrepancy in Y-axis values.

For $Q = [-1.92615834e+36, -2.14353463e+35, -2.50297148e+37, -4.11065669e+37, -9.48702371e+37]$ and $R = [-5.29541465e+37, -8.25465082e+37]$, which gives the full weightage to torque, the estimations are given by the following plots:



This solution provides slight overestimate of rotor speed but good torque estimation.

7. Conclusions

In this project, the UNSGA-III multi objective optimization technique is employed to concurrently achieve optimal torque and speed estimations for EKF-based state estimator. Unlike previous studies, where only one single objective is optimized, the proposed strategy can achieve the estimation of speed and torque directly, by optimizing the components of torque such as rotor flux and current. The multi-objective UNSGA-III tuning of EKF matrices can be used to improve the dynamic performance of the drive systems beside the state estimations. Additionally, the multi-objective method can help to investigate the system behavior in different optimal regions.

8. References

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- Using NSGA II Multiobjective Genetic Algorithm for EKF-based Estimation of Speed and Electrical Torque in AC Induction Machines I.M. Alsofyani*, N.R.N. Idris, M. Jannati, S.A. Anbaran, and Y.A Alamri UTM-PROTON Future Drive Laboratory, Power

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- Sensorless Vector Control of Induction Motor Using an EKF and SVPWM Algorithm
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Department of electrical engineering Mohamed V University, Mohammadia School of Engineers. Street Ibn Sina, B.P 765 Agdal, Rabat Morocco
- Kalman Filtering Applied to Induction Motor State Estimation, Yassine Zahraoui and Mohamed Akherraz. DOI: <http://dx.doi.org/10.5772/intechopen.92871>